5 FET/MOSFET Biasing

INTRODUCTION

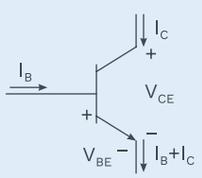
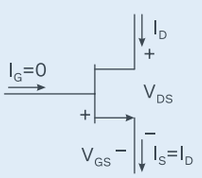
- In the previous two chapters, we learned about the bipolar junction transistor (BJT) and their application as an amplifier and switch.
- In this chapter, we will introduce a new type of transistor known as Field Effect Transistor. But the question is why do we need another type of transistor when we already have BJT? Obviously there must be some advantage of field effect transistor commonly known as FET over bipolar junction transistor.
- Besides, the differences in the properties of the two types of transistors, there exist some common aspects too. FET together with BJT constitutes three terminal devices.
- Both FET and BJT are used for amplification as well as switching purposes.
- How is the amplification of FET different from that of BJT? Before going into amplification and biasing of FET, let us discuss the important difference between BJT and FET.

Rack Your Brain

What is the advantage of FET over BJT?

Grey Matter Alert!

FET is a symmetrical device, where as BJT is an unsymmetrical device.

Comparison of BJT and FET	
BJT	**FET**
Symbol 	Symbol 
Base current is not always zero.	Gate current is always zero.
BJT is classified as npn or pnp.	FET is classified as n-channel or p-channel.
It is a bipolar device.	It is a unipolar device.

Comparison of BJT and FET	
BJT	**FET**
Terminals are known as emitter, base and collector.	Terminals are known as drain, source and gate.
It is a current controlled device.	It is a voltage controlled device.
Its operation is based on majority charge carriers as well as minority charge carriers.	Its operation depends either on majority charge carriers or minority charge carriers but not on both.
Its input impedance is low as compared to FET.	Its input impedance is very very high, that's why, gate current is always zero.
BJT is more noisy.	FET is less noisy.
Low thermal stability.	High thermal stability.
High output impedance.	Low output impedance.
It is slightly bulky.	It is less bulky as compared to BJT.
Output impedance is high.	Output impedance is low as compared to BJT.
High voltage gain.	Low voltage gain.
Less current gain.	High current gain.
Operation is slower.	Operation is faster.
Consumes high power.	Consumes less power.

Table 5.1

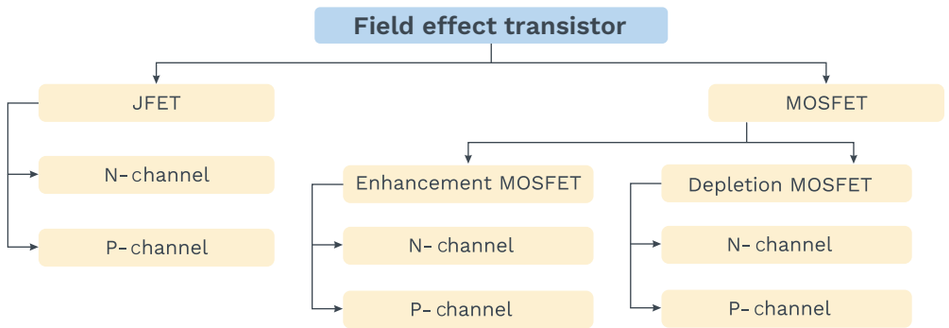
- From the above comparison, we can see that FET has various advantages over BJT namely temperature stability, input impedance etc.
- So, the obvious questions are how FET is advantageous over BJT, what are its constructional details and biasing mechanism?
- The constructional details of FET are already discussed in the subject of electronic devices and circuits. In this subject, we mainly discuss the biasing and application of FET.



Rack Your Brain

What is the reason for the high input impedance of JFET?

FET is broadly classified as-



JFET (Junction Field Effect Transistor)

As discussed already, there are two types of JFET, i.e., n-channel and p-channel, and the symbols of these are as shown below.

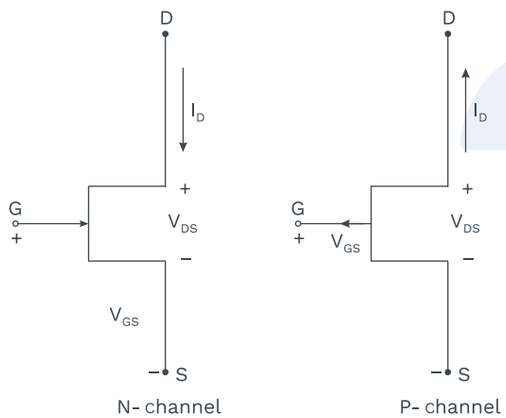


Fig. 5.1 JFET Symbols

In the n-channel JFET, current flows from drain to source, whereas in the p-channel JFET, current flows from source to drain.
A common symbol for an FET amplifier is shown below.

Grey Matter Alert!

In BJT, the output current is a function of input current where as in FET output current is a function of input voltage.

Rack Your Brain

Why is the pinch-off voltage important in JFET?

Grey Matter Alert!

Number of FET devices can be connected in parallel to increase the power and the current capacity.

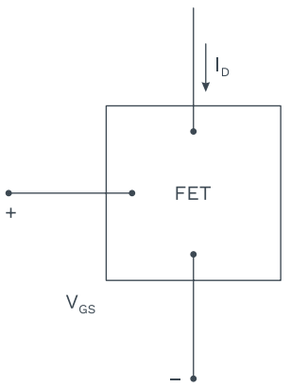


Fig. 5.2 Voltage Controlled Amplifiers Symbol

Shockley's Current Equation:

In the BJT, the current relation is linear, whereas in the FET, it is non-linear.

Drain to source current is given by

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$$

I_{DSS} = drain to source saturation current

V_{GS} = gate to source voltage which is a control variable

V_p = pinch-off voltage

For different values of V_{GS}, we have different values of drain current I_D.

When $V_{GS} = 0$, $I_D = I_{DSS}$ (maximum)

When $V_{GS} = V_p$, the drain current I_D is zero

When V_{GS} is between 0V and pinch-off voltage V_p, the current I_D will range between I_{DSS} and 0A. All the cases have been drawn in the circuit shown below.

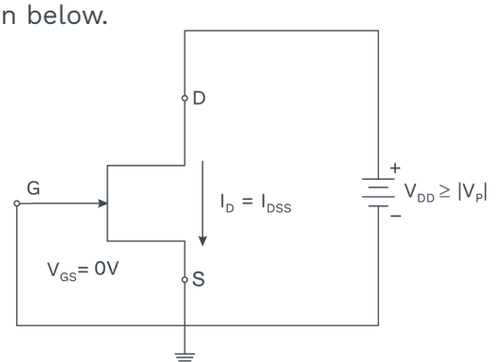


Fig. 5.3 Drain Current When V_{GS} = 0 V

Previous Years' Question

Which of the following can be considered to be the advantage of FET amplifier as compared to BJT amplifiers?

1) Higher input impedance
2) Good bias stability
3) Higher gain-bandwidth product
4) Lower noise figure

Select the correct answer using the codes given below.

Codes:

a) 1, 2 and 3
b) 1, 2 and 4
c) 2, 3 and 4
d) 1, 3 and 4

Rack Your Brain

What is the drain current if gate to source voltage V_{GS} increases further from zero to positive?

Grey Matter Alert!

Maximum current flows through the FET when gate to source voltage (V_{GS}) is zero.

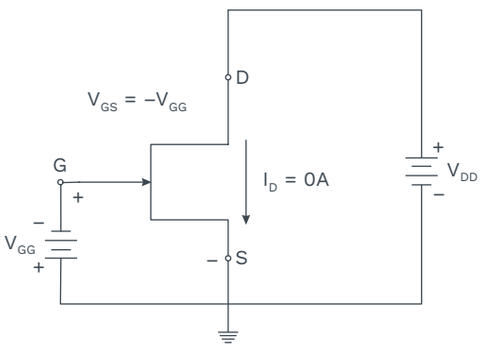


Fig. 5.4 Drain Current When $V_{GS} < V_p$

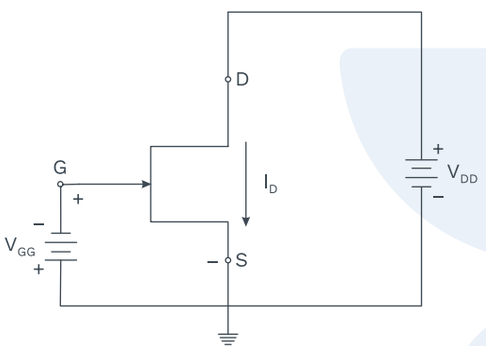


Fig. 5.5 Drain Current for $V_{GS} < 0$ V

Transfer characteristics:

Transfer characteristics can be drawn by using the information given by Shockley equation.

I_D vs V_{DS}

V_{GS} is fixed and variation of drain current I_D with drain to source voltage V_{DS} is drawn as shown below.

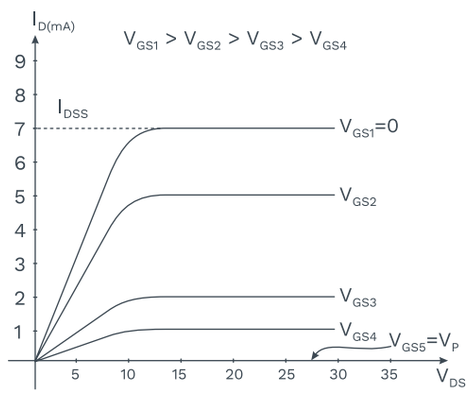


Fig. 5.6 I_D vs V_{DS} Characteristics

Previous Years' Question

The action of a JFET in its equivalent circuit can best be represented as a

a) current controlled current source.

b) current controlled voltage source

c) voltage controlled voltage source

d) voltage controlled current source

Rack Your Brain

Why FET is called as unipolar device?

Grey Matter Alert!

The gate junction is always in revere biased condition in JFET.

As the gate to source voltage increases, the drain current also increases.
When $V_{GS1} = 0$, $I_D = I_{DSS}$ (maximum)
When $V_{GS2} = V_p$, $I_D = 0$ (minimum)

I_D vs V_{GS}

Gate to source voltage V_{GS} is the controlling voltage, through which we control the drain current I_D.

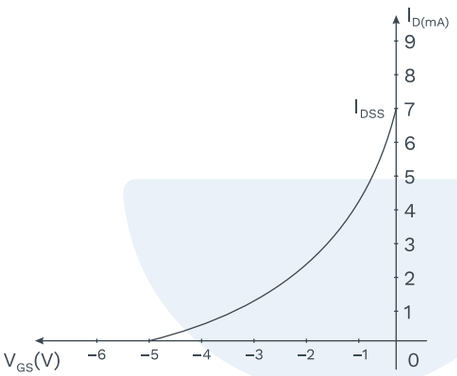


Fig. 5.7 I_D vs V_{GS} Variation

For illustration, take the pinch-off voltage to be –5 V and the drain saturation current to be 7 mA.

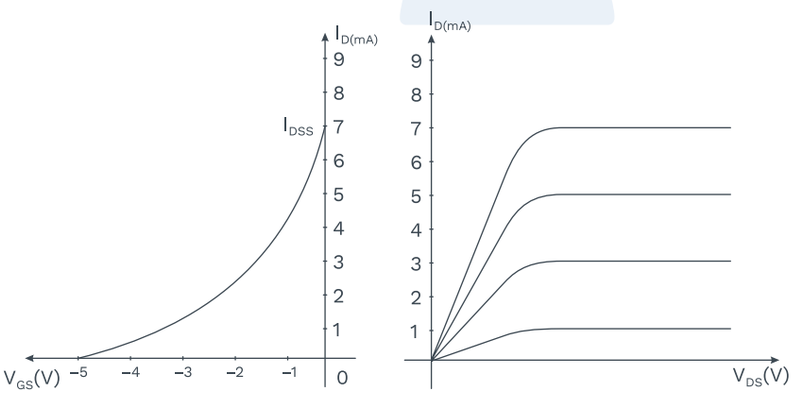


Fig. 5.8 Transfer Characteristics of JFET

BIASING CONFIGURATION OF JFET

Just like what we discussed in BJT, the same types of biasing configurations will be discussed in JFET also.

- Fixed bias configuration
- Self bias configuration
- Voltage divider Bias configuration

FIXED BIAS CIRCUIT

Fixed bias arrangement is the most simple type of biasing arrangement.
A common source fixed biased circuit is shown below.

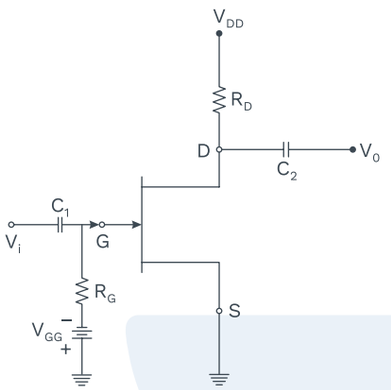


Fig. 5.9 Fixed Biased Circuits

C_1 is the capacitor used to give ac voltage, V_{DD} is for the power supply and V_{GG} is the biasing voltage.
In the above circuit, the voltage is directly applied through the gate, i.e., V_{GG}. Hence, the above circuit is known as a fixed bias circuit.

DC analysis

Replace capacitor by open circuit and short circuit the ac source.
The above circuit can be redrawn as shown below.

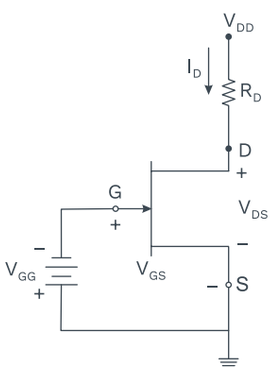


Fig. 5.10 DC Analysis of Fixed Bias Circuit

Applying KVL in the gate-source loop, we get

$$-V_{GG} - V_{GS} = 0$$

$$V_{GS} = -V_{GG}$$

Grey Matter Alert!

FET is the majority carrier device.

Drain current is given by

$$I_D = I_{DSS}\left(1-\frac{V_{GS}}{V_P}\right)^2$$

Here, V_{GS} is the fixed applied voltage and I_D depends on V_{GS}. So, I_D becomes constant, if V_P and I_{DSS} are constant.
Therefore, it is known as a fixed bias circuit.

Drain current $I_D = I_{DSS}\left(1+\frac{V_{GG}}{V_P}\right)^2$

Applying KVL in the outer loop, we get

$V_{DD} - I_D R_D - V_{DS} = 0$
$V_{DS} = V_{DD} - I_D R_D$

Example 1:
A JFET transistor has drain to source saturation current of 10 mA, V_{GS} = 4 V and pinch-off voltage of 5 V. Calculate the drain current.

Sol:
Drain current of N-channel JFET is

$$I_D = I_{DSS}\left(1-\frac{V_{GS}}{V_P}\right)^2$$

$$I_D = 10\left(1-\left(\frac{-4}{-5}\right)\right)^2$$

$$I_D = \frac{10}{25}$$

I_D = 0.4 mA

Rack Your Brain

Why fixed bias circuit always requires two dc supplies for operation?

Example 2:
What will be the drain current, if gate to source voltage is equal to V_p?

a) Maximum **b)** Minimum
c) I_{DSS} **d)** 0

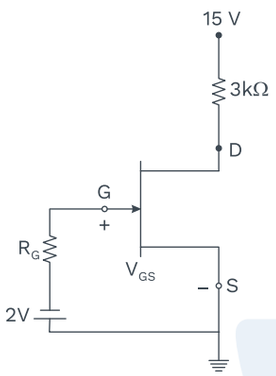
Sol:

Drain current, $I_D = I_{DSS}\left(1-\frac{V_{GS}}{V_P}\right)^2$

$V_{GS} = V_P$
I_D = 0 A
Hence, the correct option is D).

Example 3:

A fixed bias circuit is shown below, calculate the operating point, if I_{DSS} is 8 mA and $V_P = -6$ V.



Sol:

The operating point of the fixed bias circuit is I_D, V_{GS}.

Applying KVL in the input loop,

$-2 - I_G R_G - V_{GS} = 0$

$V_{GS} = -2V \quad (\because I_G = 0)$

Drain to source current is given by

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Putting all the values in the current equation

$$I_D = 8\left(1 - \left(\frac{-2}{-6}\right)\right)^2$$

$$I_D = 8\left(\frac{2}{3}\right)^2$$

$$I_D = \frac{8 \times 4}{9}$$

$I_D = 3.55$ mA

The operating point is

$I_D = 3.55$ mA

$V_{GS} = -2$ V

Example 4:

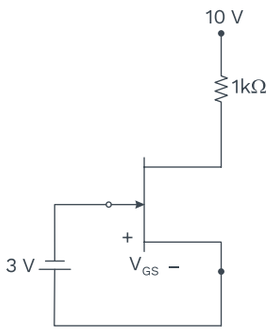
A fixed bias circuit is shown below, indicating the different voltages and resistances. Calculate V_{DS} and I_D, if $I_{DSS} = 5$ mA and $V_P = -6$ V.

Rack Your Brain

Why is the relation between drain current and gate to source voltage V_{GS} non-linear?

Grey Matter Alert!

Fixed bias and self-bias JFET configuration with a bypassed source capacitance have almost the same voltage gain.



Sol:

Applying KVL in the input loop, we get

$-3 - V_{GS} = 0$

$V_{GS} = -3V$

Drain current, $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$

$I_D = 5\left(1 - \left(\frac{-3}{-6}\right)\right)^2$

$I_D = 5 \times \frac{1}{4}$

$I_D = 1.25$ mA

Applying KVL in the outer loop, we get -

$10 - I_D \times 1 - V_{DS} = 0$

$V_{DS} = 10 - I_D$

$V_{DS} = 10 - 1.25$

$V_{DS} = 8.75$ V

SELF BIAS CONFIGURATION

In the fixed bias configuration, we always require two external voltages, V_{DD} and V_{GG}, to operate JFET.

In a self bias circuit, for biasing, we require a single power supply, i.e, V_{DD}.

The circuit for self-bias configuration is shown below.



Rack Your Brain

Which is the most economical fixed bias or self bias and why?

Grey Matter Alert!

The condition, in which the current between gate and source is blocked, is known as pinch-off condition.

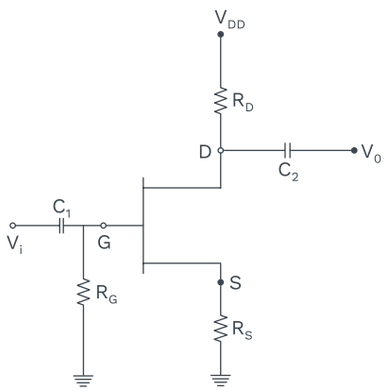


Fig. 5.11 Self-Bias Circuit

It must be noted very carefully that there is no extra supply given to the gate terminal.

DC analysis

- Replace the capacitor by an open circuit and ac voltage source by short circuit.

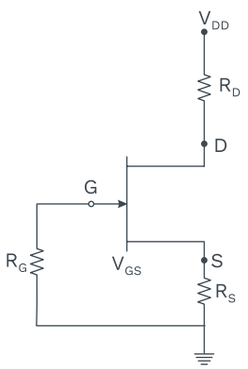
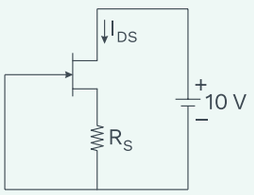


Fig. 5.12 DC Analysis of Self-Bias Circuit

- In the fixed bias circuit, there is not much use of source resistance (R_S), whereas in the case of self bias circuit, R_S is must for giving supply to the gate terminal.

Previous Years' Question

The JFET in the circuit shown in figure has an I_{DSS} = 10 mA and V_P=–5 V. The value of the resistance R_S for a drain current I_{DS} = 6.4 mA is (select the nearest value)



a) 150 ohms
b) 470 ohms
c) 560 ohms
d) 1 kilo-ohms

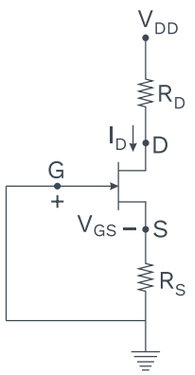


Fig. 5.13 DC Analysis of Self-bias Circuit

Since the gate current is zero, the voltage across the resistance R_G is zero. Hence, R_G is redundant in the above figure.

Why is it called as a self bias circuit?

To understand this, apply KVL in the gate-source loop, we get

$$-V_{GS} - I_D R_S = 0$$
$$V_{GS} = -I_D R_S$$

Without giving any external power supply to the gate terminal, a negative voltage is supplied. In other words, we can control V_{GS} and I_D as per the power supply. That's why, it is known as self bias circuit.

Drain current is given by

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$$

$$I_D = I_{DSS}\left(1 + \frac{I_D R_S}{V_p}\right)^2$$

Applying KVL in the outer loop, we get

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$
$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

Example 5:

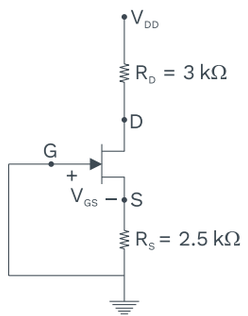
A self bias circuit is shown below with drain current of 1 mA and $V_P = -5$ V, $V_{DD} = 10$ V, find V_{GS} and V_{DS}.

Rack Your Brain

How does the voltage divider circuit get gate biasing voltage from a single source?

Grey Matter Alert!

The order of leakage current in MOS is in nanoamperes.

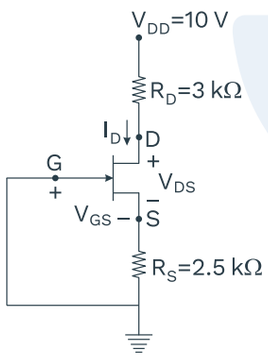


Grey Matter Alert!

Voltage divider bias is the most stable circuit from the biasing point of view.

Sol:

The given circuit can be redrawn as shown below.



Applying KVL in the inner loop, we get

$V_{GS} = -I_D R_S$

$R_S = 2.5\ k\Omega$

$I_D = 1\ mA$

$V_{GS} = -1 \times 2.5$

$V_{GS} = -2.5\ V$

Applying KVL in the outer loop, we get

$10 - 3I_D - V_{DS} - 2.5I_D = 0$

$V_{DS} = 10 - 5.5I_D$

$V_{DS} = 10 - 5.5 \times 1$

$V_{DS} = 4.5\ V$

Example 6:

The limitation of using a fixed bias circuit is

a) It requires external power supply

b) It requires external resistance

c) It has high-temperature sensitivity

d) None of the above

Sol:
The limitation of using a fixed bias circuit is that we have to supply negative bias voltage to the gate terminal.
So, it requires two power supplies.
Hence, the correct option is A).

VOLTAGE DIVIDER BIASING

The voltage divider circuit for FET is the same as that of BJT. In the voltage divider biasing, two resistances R_1 and R_2 are used.
The circuit for voltage divider biasing is shown below-

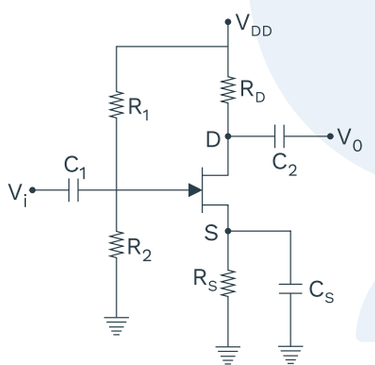


Fig. 5.14 Voltage Divider Biasing Circuit

Voltage to the gate is supplied through the power supply V_{DD} using the voltage divider rule. So in this circuit also, we require a single power supply voltage V_{DD} for operation.



Fig. 5.15 Voltage Division Circuit

Rack Your Brain

Why transconductance is the ratio of I_D with V_{GS} and not V_{DS}?

Grey Matter Alert!

The steeper the slope of I_D vs V_{GS} curve, the greater is the level of g_m.

Previous Years' Question

An n-channel JFET has I_{DSS}= 1 mA and V_P = –5 V. Its maximum transconductance (in mS) is___.

Using the voltage division rule, Thevenin's equivalent voltage is that

$$V_{R_2} = \frac{V_{DD}R_2}{R_1 + R_2}$$

Thevenin's equivalent resistance is that

$$R_{Th} = \frac{R_1R_2}{R_1 + R_2}$$

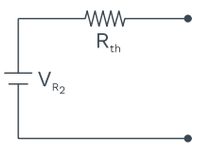


Fig. 5.16 Thevenin's Equivalent Circuit

Rack Your Brain

Why should we go for MOSFET, if we already have JFET?

DC analysis

For dc analysis, replace the capacitor by open circuit and input ac source by short circuit.
The overall circuit using voltage division and Thevenin's equivalent circuit is shown below

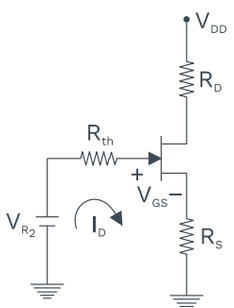


Fig. 5.17 Voltage Divider Simplifier Circuit

Grey Matter Alert!

MOSFETs are one of the principal elements in the manufacturing of ICs.

Applying KVL in the inner loop, we get

$V_{R2} - 0 \times R_{Th} - V_{GS} - I_DR_S = 0$

$V_{GS} = V_{R2} - I_DR_S$

Applying KVL in the outer loop, we get

$V_{DD} - I_DR_D - V_{DS} - I_DR_S = 0,$

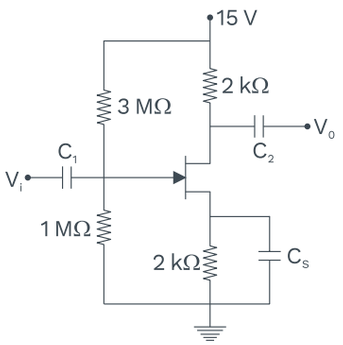
$V_{DS} = V_{DD} - I_D(R_D + R_S)$

Grey Matter Alert!

Why only a layer of silicon dioxide SiO_2 is used in the implementation of MOSFET?

Example 7:

A voltage divider bias circuit is shown below with I_{DSS} = 6 mA and V_P = –4 V and drain current, I_D = 1 mA.



Calculate

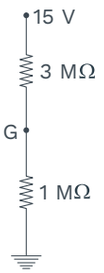
a) V_{GS} **b)** V_{DS}

c) V_S **d)** V_D

Sol:

DC analysis

For dc analysis, replace the capacitor by open circuit and calculate Thevenin's equivalent of the voltage divider circuit.



$$V_G = \frac{15 \times 1}{1+3}$$

$$V_G = 3.75 \text{ V}$$

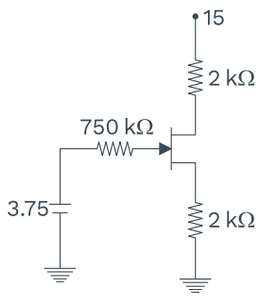
Thevenin's equivalent resistance,

$$R_{Th} = \frac{3 \times 1}{3+1} = 0.75 \text{ MΩ} = 750 \text{ kΩ}$$

DC circuit can be redrawn as shown below.

Grey Matter Alert!

MOSFET is faster than BJT mainly because it is a unipolar device, i.e., and it doesn't have any charge storage problem.



Applying KVL in the input loop, we get

$3.75 - V_{GS} - 2 = 0$

$V_{GS} = 1.75$ V

Applying KVL in the outer loop, we get

$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$

$V_{DS} = V_{DD} - I_D(R_S + R_D)$

$V_{DS} = 15 - 1 \times 4$

$V_{DS} = 11$ V

$V_S = I_D R_S$

$V_S = 1 \times 2$

$V_S = 2$ V

$V_D = V_{DD} - I_D R_D$

$= 15 - 1 \times 2$

$V_D = 13$ V

Example 8:

The pinch-off voltage for an n-channel JFET is 6V, when $V_{GS} = -2$ V, the pinch-off occurs for V_{DS} is

a) 1 V **b)** 2 V

c) 3 V **d)** 4 V

Sol:

Pinch-off voltage $V_{P1} = 6$ V

For $V_{GS} = -2$ V

$V_{P2} = ?$

$V_{P2} = V_{DS} = V_P - |V_{GS}|$

$V_{DS} = 6 - 2$

$V_{DS} = 4$ V

Hence, the correct option is d).

TRANSCONDUCTANCE OF JFET

Transconductance is defined as the rate of variation of drain current with respect to the gate to source voltage.

Grey Matter Alert!

BJT will have input impedance in the range of 1–3kΩ whereas MOSFET will have input impedance in the range of 10^{10} to 10^{16} Ω.

It is a very important parameter, because in MOSFET, we control the output drain current using gate to source voltage V_{GS}, and transconductance parameter gives us an indication of the same.
It is denoted by g_m and is given as,

$$g_m = \frac{\partial I_d}{\partial V_{GS}}$$

Drain current, $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$

Differentiate the above equation partially with respect to V_{GS},

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{2I_{DSS}}{-V_P}\left(1 - \frac{V_{GS}}{V_P}\right)$$

$$|g_m| = \frac{2I_{DSS}}{V_P}\left(1 - \frac{V_{GS}}{V_P}\right)$$

MOSFET (Metal oxide semiconductor Field Effect Transistor)

- The next type of FET that we will study is MOSFET. In MOSFET, there exist three terminals, i.e., drain, gate and source.
- In MOSFET, the current between drain and source is controlled by a third terminal that is gate.
- By controlling the gate voltage, the width of the channel is controlled and the gate is isolated from the device using an insulator SiO_2.
- The important question is "why should we use MOSFET when we already have an amplifier JFET?". Obviously, there are various advantages with MOSFET that are not with JFET.
- Before going into the depth of MOSFET, it is important to compare MOSFET and JFET, and see how the properties of the MOSFET could be considered different from that of the JFET?

Rack Your Brain

What is the width to length ratio called?

Grey Matter Alert!

Pinch-off voltage is the gate to source voltage at which the drain current is zero.

Rack Your Brain

How the performance of E-MOSFET is different from that of D-MOSFET?

JFET	MOSFET
It operates in signal mode.	It operates in depletion mode and enhancement mode.
It has relatively less input impedance.	It has relatively very high input impedance.
Leakage current is in the range of nano ampere.	Leakage current is in the range of pico ampere.
Used for low noise application.	Useful for high noise application.
Relatively cheaper.	Relatively expensive.

Table 5.2

General construction

The constructional details of MOSFET are already discussed in the subject of electronic devices and circuits.

Let us quickly review the construction details and working.

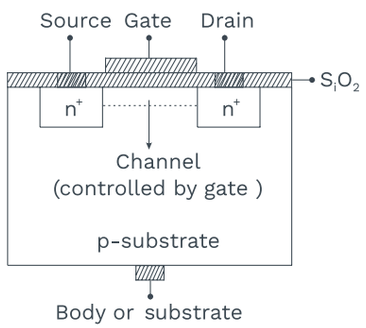


Fig. 5.18 MOSFET Construction

- A p-type or n-type substrate is taken, which should not be highly doped.
- On this p-substrate or n-substrate, a heavily doped n-type or p-type material is formed.
- A very thin oxide layer is formed known as the SiO_2 layer.
- Three terminals are known as source, gate and drain and are taken through the aluminium metal contact.

Grey Matter Alert!

Depletion type MOSFET is called as normally ON device.

Rack Your Brain

What are the different modes of operation of an E-MOSFET?

Grey Matter Alert!

Channel length modulation in MOSFET is equivalent to early effect in BJT.

- A fourth terminal is taken from the other side of the substrate.
- It can be clearly seen that three terminals are isolated from the heavily doped substrate by the thin oxide layer. That is why a MOSFET is also known as an IGFET (insulated gate field effect transistor).

Channel

The channel shown in the construction plays a very important role. Sometimes, the channel is created and sometimes not.

It depends on whether the channel is present or not, MOSFET is classified as a depletion type or an enhancement type.

DEPLETION TYPE

In the depletion type MOSFET, a channel is present between source and drain as shown below.

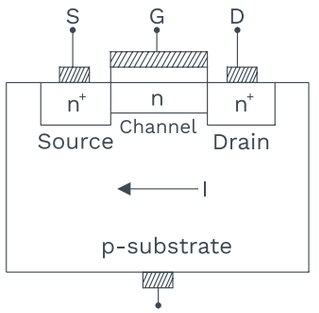


Fig. 5.19 Depletion Type MOSFET

Important points

- It has SiO_2 layer, which has a width of 0.2–0.1 μm.
- Due to the presence of an insulator at the gate terminal, the input impedance is very high.
- It is a planar device.
- The area of MOSFET is 5% of BJT.
- MOSFET is a symmetrical device, so the source and drain can be interchanged.
- MOSFET is a four terminal device.
- MOSFET can also work as a capacitor and is called as a MOS capacitor.

Rack Your Brain

In which region, MOSFET is used as an amplifier?

Grey Matter Alert!

The threshold voltage is the minimum gate to source voltage, above which MOSFET goes into the saturation region.

Working

- The voltage applied between gate and source controls the channel, and we can say that it can control channel resistance between the source and drain.
- The actual charge flow between the source and drain depends on the voltage applied between the drain and source.

Case 1: When V_{GS} = 0 V and V_{DS} = 0 V
Channel is present but without V_{DS}, the current cannot flow through the channel.

I_D = 0 A

Case 2: When V_{GS} = 0 V and V_{DS} > 0 V
Channel is present and the voltage is applied across the channel; thus, current flows from drain to source.

Case 3: When V_{DS} > 0 V and V_{GS} < 0 V
An electric field is produced pointing towards gate terminal, and it pushes electron downward or pushes electron away from the channel.

Characteristics

I_D vs V_{DS}

The variation of I_D with respect to V_{DS} is shown below.

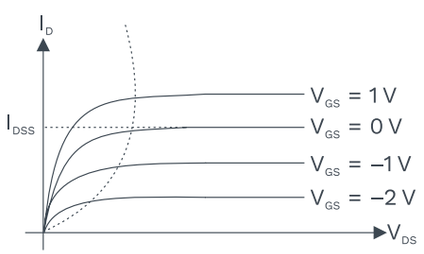


Fig. 5.20 I_D vs V_{DS} Characteristics

As V_{DS} increases, drain current I_D also increases, and it reaches maximum at V_{GS} = 0V.
At some particular value of V_{GS}, I_D becomes saturated.
I_D vs. V_{GS} graph is shown below.

Rack Your Brain

What are the different biasing techniques in MOSFET?

Grey Matter Alert!

RF input impedance of MOSFET is low due to interelectrode capacitance.

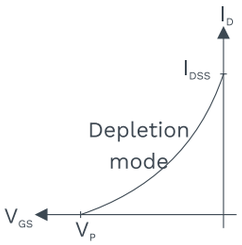


Fig. 5.21 I_D vs V_{GS}

Representation of D-MOSFET is

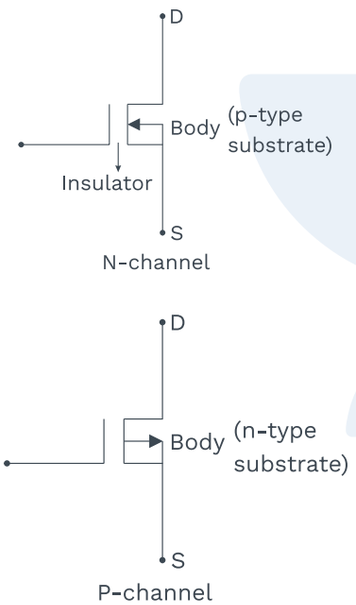


Fig. 5.22 Symbol of MOSFET

REGIONS OF OPERATION

1) Triode region

$V_{DS} \leq V_{GS} - |V_T|$

Where V_T = threshold voltage

$$I_D = 2K\left[(V_{GS} - |V_T|)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

$$K = \mu_n C_{ox}\frac{W}{2L}$$

C_{ox} = oxide capacitance $= \frac{\epsilon_{ox}}{t_{ox}}\frac{F}{m^2}$

Rack Your Brain

Which is the most stable biasing circuit among all the configurations?

Previous Years' Question

An n-channel depletion MOSFET has the following two points on its I_D–V_{GS} curve:

i) $V_{GS} = 0$ at $I_D = 12$ mA

ii) $V_{GS} = -6$ at $Z_0 = \infty$

Which of the following Q-points will give the highest trans-conductance gain for small-signals?

a) $V_{GS} = -6$

b) $V_{GS} = -3$

c) $V_{GS} = 0$

d) $V_{GS} = 3$

W = channel width

L = channel length

$\frac{W}{L}$ = aspect ratio

2) Saturation region

$V_{DS} > V_{GS} - |V_T|$

$I_D = K(V_{GS} - |V_T|)^2$

If $V_{GS} = 0V$

$I_D = K|V_T|^2 = I_{DSS}$

or $K = \frac{I_{DSS}}{|V_T|^2}$

3) Cut-off Region

$V_{GS} < V_P$ (Pinch-off voltage)

- If V_{GS} is more negative than pinch-off voltage, then $I_D = 0$ A.
- For p-channel MOSFET, we have to reverse the polarity of each voltage.

  $V_{DS} \rightarrow V_{SD}$

  $V_{GS} \rightarrow V_{SG}$

ENHANCEMENT TYPE MOSFET

- In an enhancement type MOSFET, the channel is not present.
- Enhancement MOSFET is easy to fabricate, because we do not need to create the channel during fabrication.

Constructional details

The construction of enhancement type is very similar to that of depletion, except when the channel is absent.

Rack Your Brain

What is the difference between BJT and MOSFET current mirror circuits?

Grey Matter Alert!

In the voltage divider biasing, we can always assume the gate current to be 0 A to permit an isolation from the output.

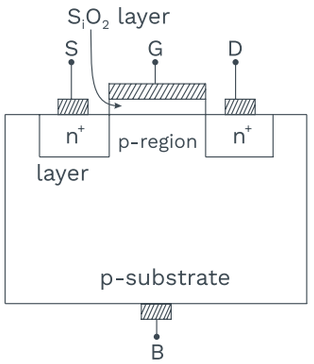


Fig. 5.23 Enhancement Type MOSFET

Region of operation

There are namely 3 regions of operation of enhancement type MOSFET.

- Accumulation mode
- Depletion mode
- Inversion mode

Accumulation mode

In the accumulation region, holes get accumulated below the gate region, and electrons are pushed away from the channel region.

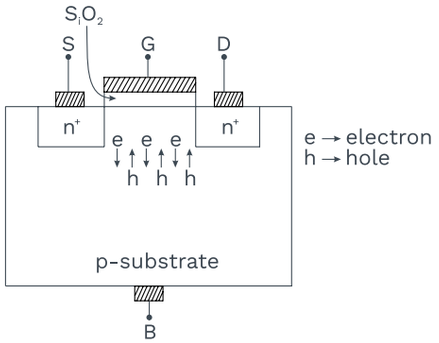


Fig. 5.24 Accumulation Mode

Depletion mode

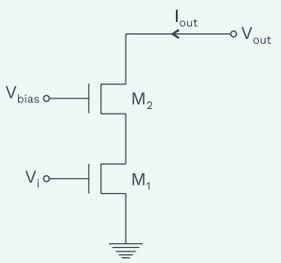
In depletion mode, the region below the gate terminal becomes depleted of holes, but electrons start to accumulate below the gate terminal.

Grey Matter Alert!

MOSFET has more switching speed than BJT.

Previous Years' Question

Two identical NMOS transistors M_1 and M_2 are connected as shown below. V_{bias} is chosen so that both transistors are in saturation. The equivalent g_m of the pair is defined to be $\frac{\partial I_{out}}{\partial V_i}$ at constant V_{out}.



The equivalent g_m of the pair is

a) The sum of individual g_m's of the transistors

b) The product of individual g_m's of the transistors

c) Nearly equal to the g_m of M_1

d) Nearly equal to g_m/g_0 of M_2

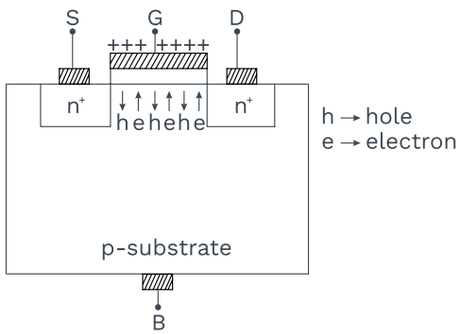


Fig. 5.25 Depletion Mode

Rack Your Brain

What is the difference between g_d and g_m?

Inversion mode

- Since we continuously remove holes and accumulate electrons such that after a certain point, electron concentration > hole concentration below gate terminal.
- The nature of that region becomes n-type. Hence, the nature of the material is inverted.
- In this case, a channel is formed between source and drain.
- For this condition, V_{DS} must be greater than a certain value called as threshold voltage.

Representation of E-MOSFET is

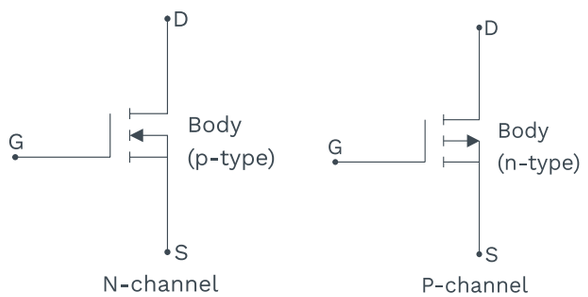


Fig. 5.26 Symbols of E-MOSFET

Grey Matter Alert!

The current mirror circuit is used to power up multiple transistors using a single power supply.

Operating regions of E-MOSFET

1) Cut-off region

$V_{GS} < V_T$

No channel is present between source and drain.

So, $I_D = 0$ A

2) Linear or Triode region,

$V_{GS} > V_T$, $V_{DS} < V_{GS} - V_T$

$$I_D = 2K\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

3) Saturation region,

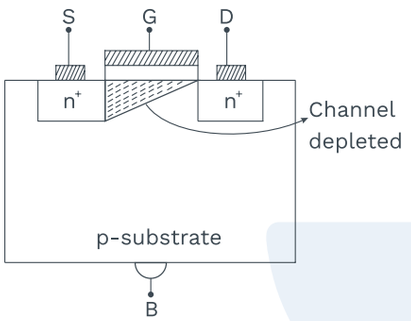


Fig. 5.27 E MOSFET in Saturation

$V_{DS} \geq V_{GS} - V_T$

$I_D = K[(V_{GS} - V_T)]^2$

Channel length modulation

- When drain voltage is increased, and it dominates gate voltage, the p-n junction of drain and substrate becomes reverse biased. Hence, the depletion layer grows near the drain terminal.
- The channel gets depleted near drain terminal and effective channel length reduces.

  $I_D \propto \frac{W}{L}$ and hence, I_D increases.

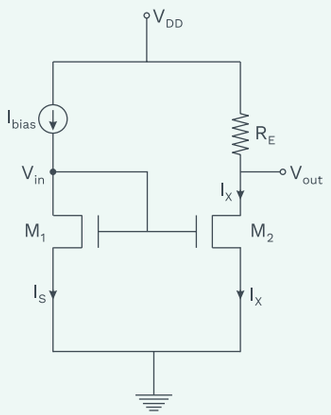
MOSFET BIASING

Similar to JFET and BJT, MOSFET also requires dc biasing voltage to operate as an amplifier. In accordance with the biasing circuits that we studied for JFET and BJT, similar circuits are observed for MOSFET that are-

- Fixed bias configuration
- Drain to gate feedback resistance
- Voltage divider biasing
- Constant current source

Previous Years' Question

For the circuit shown in the following figure, transistors M_1 and M_2 are identical NMOS transistors. Assume that M_2 is in saturation, and the output is unloaded.



The current I_X is related to I_{bias} as

a) $I_X = I_{bias} + I_S$

b) $I_X = I_{bias}$

c) $I_X = I_{bias} - I_S$

d) $I_X = I_{bias} - \left(V_{DD} - \frac{V_{out}}{R_E}\right)$

Rack Your Brain

How channel length modulation is different from the early effect?

1) Fixed bias configuration

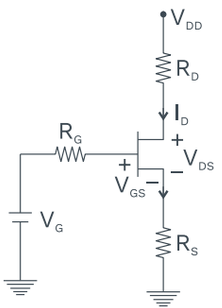


Fig. 5.28 MOSFET fixed bias circuit

Applying KVL in the input loop, we get

$$V_G - I_G R_G - V_{GS} - I_D R_S = 0$$

$$V_{GS} = V_G - I_D R_S \qquad (I_G = 0 \text{ A})$$

Applying KVL in the output loop, we get

$$V_{DS} = V_{DD} - I_D R_D - I_D R_S$$

Assume MOSFET operates in a saturation region:

$$I_D = K(V_{GS} - V_T)^2$$

$$I_D = K(V_G - I_D R_S - V_T)^2$$

$$= K(V_G - I_D R_S - V_T)^2$$

Complete $V_{GS} = V_G - I_D R_S$ and

$$V_{DS} = V_{DD} - I_D R_D - I_D R_S$$

- If $V_{DS} \geq V_{GS} - V_T$

Then our assumption was correct, and MOSFET operates in the saturation region.

- If $V_{DS} < V_{GS} - V_T$

Then our assumption was wrong, and MOSFET operates in the linear region.

$$I_D = 2K\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

Grey Matter Alert!

BJT has more transconductance than FET.

Rack Your Brain

What is the typical value of aspect ratio of MOSFET?

Rack Your Brain

What is the difference between the small-signal analysis of FET and BJT?

2) Drain to gate feedback resistance

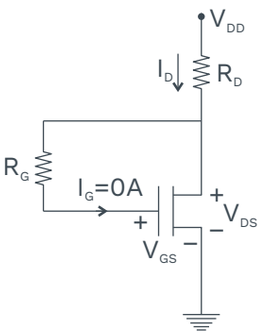


Fig. 5.29 Drain to Gate Feedback

- It is similar to collector base bias in BJT.
- Since $I_G = 0$ A, $V_G = V_D$

MOSFET operates in the saturation region

$V_D = V_{DD} - I_D R_D = V_G$

$V_{GS} = V_G - V_S = V_G$

$I_D = K(V_{GS} - V_T)^2 = K(V_G - V_T)^2$

$I_D = K(V_{DD} - I_D R_D - V_T)^2$

Grey Matter Alert!

There is zero phase shift between input and output in source follower and common gate configurations.

3) Voltage divider biasing

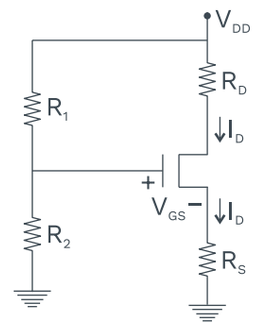


Fig. 5.30 MOSFET Voltage Divider Biasing

- As in the case of BJT, there is no need to compute Thevenin's equivalent, because I_G= 0 A. So R_1 and R_2 are always in series.

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2}$$

$V_S = I_D R_S$

Grey Matter Alert!

The gain of FET is typically between 2 and 20.

$$V_{GS} = V_G - V_S$$

$$V_{GS} = \frac{V_{DD}R_2}{R_1 + R_2} - I_D R_S$$

Assume MOSFET operates in the saturation region.

So, $I_D = K(V_{GS} - V_T)^2$

$= K(V_G - V_S - V_T)^2$

Solve the quadratic equation to determine I_D,

Check $V_{DS} = V_{DD} - I_D R_D - I_D R_S$

If $V_{DS} \geq V_{GS} - V_T$: Saturation (assumption correct)

If $V_{DS} < V_{GS} - V_T$: Linear

$$I_D = 2K\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

4) Constant current source

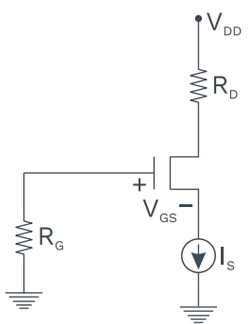


Fig. 5.31 MOSFET Constant Current Source

Drain current is fixed at I_S,

Since, $I_G = 0$ A, $V_G = 0$ V

$V_D = V_{DD} - I_D R_D$

$V_D = V_{DD} - I_S R_D$

MOSFET CURRENT MIRROR

In BJT, we studied about current mirror circuit and its application as a constant current source. In BJT, we connect the base and emitter of two transistors together.

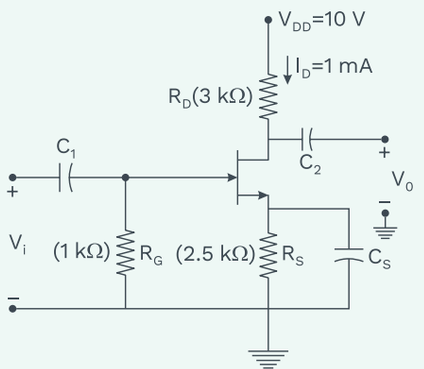
In MOSFET, we connect the gate and source of transistors together.

Rack Your Brain

What is the role of coupling capacitors in an amplifier circuit?

Previous Years' Question

The voltage gain $A_v = \frac{V_0}{V_i}$ of the JFET amplifier shown in the figure is



I_{DSS} = 10 mA, V_P = −5 V (assume C_1, C_2 and C_S to be very large)

a) +18 **b)** −18

c) +6 **d)** −6

- In MOSFET, gate current = 0, $I_G = 0$ A

Since V_{gs} is same for both transistors and both transistors are identical.

$I_{D1} = I_{D2}$

$I_0 = I_{ref}$

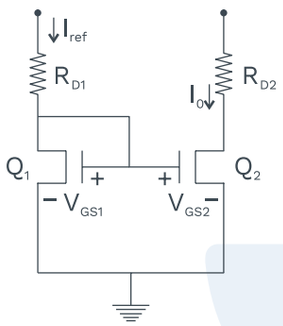


Fig. 5.32 MOSFET Current Mirror Circuit

If the aspect ratio of both transistors are not equal then,

$$\frac{I_0}{I_{ref}} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

$$\because I_D = \mu_n C_{ox} \frac{W}{2L}(V_{GS} - V_T)^2$$

V_{GS} and V_T are same for both transistors.

In BJT, currents were scaled on the basis of device area and in MOSFET, we use aspect ratio.

MOSFET CURRENT STEERING CIRCUIT

The current steering circuit is an application of the MOS current source, and it is used to implement constant current biasing.

Q_1 current mirror exists with Q_2 and Q_3

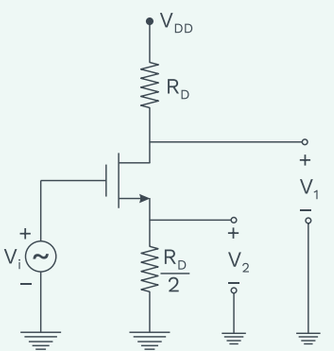
$$\frac{I_2}{I_{ref}} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

Grey Matter Alert!

Common gate configuration has the lowest input impedance among all the configurations of FET.

Previous Years' Question

In the MOSFET amplifier of the figure, the signal output V_1 and V_2 obey which of the following relationship.



a) $V_1 = \frac{V_2}{2}$

b) $V_1 = -\frac{V_2}{2}$

c) $V_1 = 2V_2$

d) $V_1 = -2V_2$

$$\frac{I_3}{I_{ref}} = \frac{\left(\frac{W}{L}\right)_3}{\left(\frac{W}{L}\right)_1}$$

$I_4 = I_3$: acts as a reference current for other current mirror circuits.

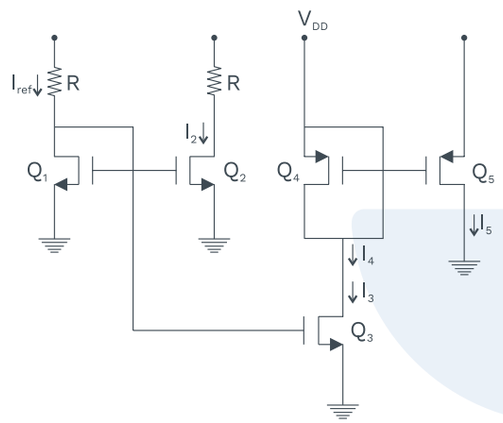


Fig. 5.32 MOSFET Current Steering Circuit

Channel conductance,

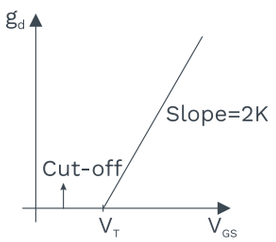


Fig. 5.34 g_d Varies Linearly with V_{GS}

In the linear region,

$$I_D = 2K\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

If V_{DS} is small, then $\frac{V_{DS}^2}{2}$ is ignored.

$$I_D = 2K[(V_{GS} - V_T)V_{DS}]$$

$$g_d = \frac{\partial I_D}{\partial V_{DS}} = 2K(V_{GS} - V_T)$$

Rack Your Brain

What is the similarity between source follower and emitter follower circuit?

Grey Matter Alert!

Except source follower configuration, the output impedance of all FET configurations is determined by R_D.

Rack Your Brain

What is MESFET and how it is different from that of MOSFET?

$$r_d = \frac{1}{g_d} = \text{channel resistance.}$$

Transconductance of MOSFET

It is denoted by g_m, and is given as

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

Just like BJT acts as an amplifier in the active region, and MOSFET acts as an amplifier in the saturation region.

$I_D = K(V_{GS} - V_T)^2$

$$\Rightarrow \quad g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = 2K(V_{GS} - V_T)$$

$$g_m = 2K\sqrt{\frac{I_D}{K}} = 2\sqrt{KI_D} \qquad \left\{V_{GS} - V_T = \sqrt{\frac{I_D}{K}}\right\}$$

This g_m is used in a small-signal model.

- To determine g_m, we need to determine dc bias current I_D.

  $$g_m = 2\sqrt{KI_D}$$

- Practically in saturation,

  $I_D = K(V_{GS} - V_T)^2(1 + \lambda V_{DS})$

  $$\frac{\partial I_D}{\partial V_{DS}} = K(V_{GS} - V_T)^2\lambda$$

  $$r_d = \frac{1}{\left(\frac{\partial I_D}{\partial V_{DS}}\right)} = \frac{1}{K(V_{GS} - V_T)^2\lambda}$$

If channel length modulation is ignored, $\lambda = 0$

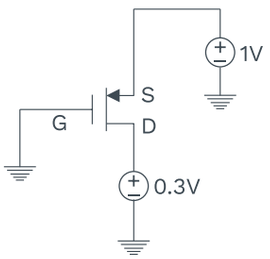
$\therefore r_d \longrightarrow \infty$

Example 10:

For the circuit shown below. If V_{Th} = 0.4 V, then find the region of transistor.

Grey Matter Alert!

The ac analysis of JFET and D-MOSFET is same.



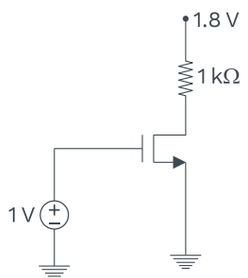
Sol:

$V_S = 1$ V, $V_G = 0$ V
$V_D = 0.3$ V, $V_{SG} = 1$ V
$V_{SD} = 0.7$ V
$V_{SD} > V_{SG} - |V_T|$
$0.7 > 1 - 0.4$
$0.7 > 0.6$

Hence, the transistor operates in saturation region.

Example 11:

In a given circuit $V_{Th} = 0.4$ V, $\mu_n C_{ox} = 200\ \mu A/V^2$. If the transistor operates at edge of saturation, then W/L is _____.



Sol:

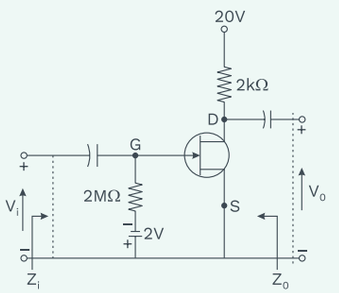
$V_G = 1$ V
$V_S = 0$ V
$V_{GS} = 1$ V at edge of saturation
$V_{DS} = V_{GS} - V_T = 1 - 0.4$
$= 0.6$ V

$$I_D = \frac{V_{DD} - V_{DS}}{R_D}$$

$$I_D = \frac{1.8 - 0.6}{1K} = 1.2 \text{ mA}$$

Previous Years' Question

Consider JFET amplifier shown in below figure,



Given, $r_d = 20$ kΩ, $I_{DSS} = 10$ mA, $V_p = -8$ V

i) Z_i and Z_0 of the circuit are respectively

a) 2 MΩ and 2 kΩ
b) 2 MΩ and 20/11 kΩ
c) Infinity and 2 MΩ
d) Infinity and 20/11 MΩ

ii) I_D and V_{DS} under DC condition are respectively

a) 5.625 mA and 8.75 V
b) 7.500 mA and 5.00 V
c) 4.500 mA and 11.00 V
d) 6.250 mA and 7.50 V

iii) Transconductance in milli-Siemens (mS) and voltage gain of the amplifier are respectively.

a) 1.875 mS and 3.41
b) 1.875 mS and −3.41
c) 3.3 mS and −6
d) 3.3 mS and 6

$$1.2\text{mA} = \mu_n C_{ox} \frac{W}{2L}(V_{GS} - V_T)^2$$

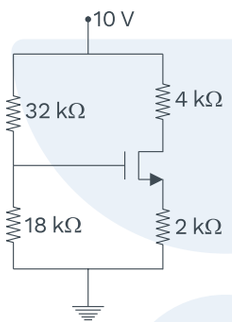
$$1.2 \times 10^{-3} = 0.2 \times 10^{-3} \frac{W}{2L}(0.6)^2$$

$$\frac{W}{2L} = \frac{1.2}{0.072}$$

$$\left(\frac{W}{L}\right) = \frac{1200}{36} = 33.33$$

Example 12:

For the circuit shown below, if V_{Th} = 0.8 V, K = 0.5 mA/V^2,



Find (i) V_{GS} and (ii) I_D

Sol:

$$V_G = \frac{10 \times 18}{18 + 32} = 3.6 \text{ V}$$

$V_S = I_D \times 2$

$V_{GS} = 3.6 - 2I_D$

Assume saturation region

$I_D = 0.5(3.6 - 2I_D - 0.8)^2$

$I_D = 0.5 \times 4(1.4 - I_D)^2 = 2[1.96 + I_D^2 - 2.8I_D]$

$I_D^2 - 3.3I_D + 1.96 = 0$

I_D = 0.776, 2.523 mA

$V_{GS} = 3.6 - 2I_D = 3.6 - 2\times(0.776) = 2.048\text{V}$

or

$V_{GS} = 3.6 - 2 \times 2.523 < 0$ (Invalid)

By KVL, $V_{DS} = 10 - 4I_D - 2I_D$

$= 10 - 6I_D$

V_{DS} = 5.344 V

$V_{DS} > V_{GS} - V_{Th}$

5.344 > 2.046 – 0.8

Hence, our assumption is correct.

SMALL-SIGNAL MODELS OF MOSFET

There are mainly 3 configurations, in which MOSFET is used

- Common source amplifiers
- Common drain amplifiers
- Common gate amplifiers

In this section, we will discuss about small-signal equivalent circuit of each of the given amplifiers.

COMMON SOURCE AMPLIFIERS

- It is similar to the common emitter amplifier of BJT.
- In a common source amplifier, input signal is applied to gate terminal, and output is taken across drain terminal.
- Input ac signal is applied through a coupling capacitor C, connected as shown below.

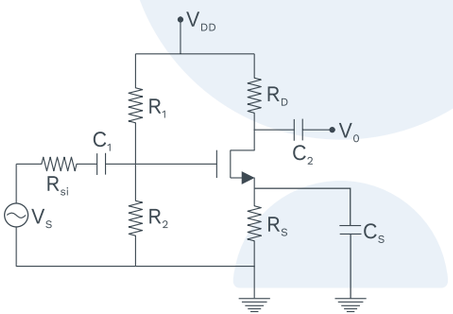


Fig. 5.35 Common Source Amplifier Circuit

- Similar to BJT amplifier, if we ground source terminal through a resistance R_S, then that resistance will act as a negative feedback.
- Due to presence R_S, gain reduces.

DC analysis

- Replace the capacitor by open circuit.
- AC voltage source replaced by short circuit.
- Since this is a voltage divider bias and I_G = 0 A, R_1 and R_2 are in series.

$$V_G = \frac{V_{DD}R_2}{R_1 + R_2}, \; V_S = 0$$

$$V_{GS} = V_G$$

Drain current, $I_D = \mu_n C_{ox} \frac{W}{2L}(V_{GS} - V_T)^2$

$$g_m = 2\sqrt{KI_D} \qquad \left(K = \mu_n C_{ox} \frac{W}{2L}\right)$$

AC analysis

- Replace capacitors by short circuit.
- DC voltage source replaced by short circuit.
- R_1 and R_2 are now in parallel, and are represented by its Thevenin's equivalent resistance.

$$R = R_1 \parallel R_2$$

$$R = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

Since source terminal is grounded, then R is connected between gate and ground.

The small-signal model is shown below.

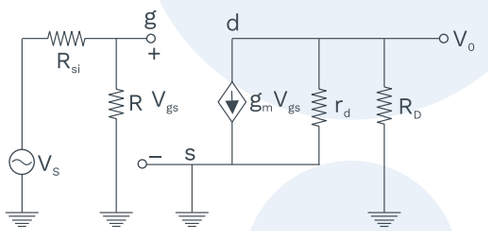


Fig. 5.36 Small-Signal Model of CS Amplifier

r_d is same as r_0 in BJT.

$$I_D = \mu_n C_{ox} \frac{W}{2L}(V_{gs} - V_T)^2(1 + \lambda V_{ds})$$

$$r_d = \frac{1}{\left(\frac{\partial I_d}{\partial V_{ds}}\right)} = \left(\mu_n C_{ox} \frac{W}{2L}\left(V_{gs} - V_T\right)^2 \lambda\right)^{-1}$$

By voltage division rule, we get

$$V_{gs} = \frac{V_S R}{R_{Si} + R}$$

$$V_0 = -g_m V_{gs}(r_d \parallel R_D)$$

Substitute V_{gs} in V_0 expression

$$V_0 = -g_m \frac{R(r_d \parallel R_D)V_s}{R_{Si} + R}$$

Voltage gain, $A_V = \frac{V_0}{V_S}$

$$A_V = -g_m \frac{(R_1 \,||\, R_2)(r_d \,||\, R_D)}{R_{Si} + (R_1 \,||\, R_2)}$$

Input resistance

$$R_i = R_1 || R_2$$

Output resistance

$$R_{out} = r_d || R_D$$

COMMON SOURCE AMPLIFIER WITH SOURCE RESISTANCE

The CS amplifier with source resistance is shown in the figure below.

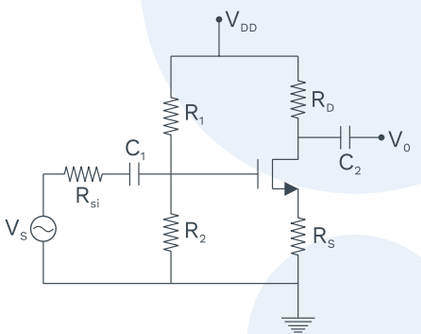


Fig. 5.37 CS Amplifier with R_S

- Due to resistance R_S, gain will get reduced.

DC analysis

- Replace capacitor by open circuit and ac input voltage by short circuit.
  Using voltage division rule-

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2}$$

$$V_S = I_D R_S$$

$$V_{GS} = V_G - V_S$$

In saturation region-

$$I_D = \mu_n C_{ox} \frac{W}{2L} (V_{GS} - V_T)^2$$

Solve quadratic to find the drain current I_D.

AC analysis

- Replace capacitors by short circuit.
- DC source will have to be replaced by short circuit. The small-signal model is shown below.

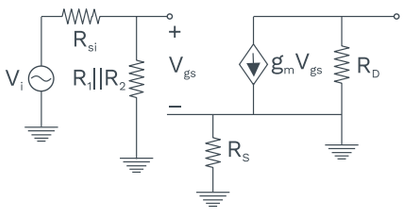


Fig. 5.38 CS Amplifier with R_S

- Ignore r_d, because it is very high. Also, it simplifies analysis.
  Input resistance:
  $R_{in} = R_1 \parallel R_2$
  Output resistance:
  $R_0 = R_D$
  Voltage gain:
  Voltage across $R_1 \parallel R_2$ = voltage between gate and ground

$$\frac{V_S(R_1 \parallel R_2)}{R_{Si} + (R_1 \parallel R_2)} = V_{gs} + g_m V_{gs} R_S$$

$$V_{gs} = V_S \left[\frac{R_1 \parallel R_2}{R_{si} + R_1 \parallel R_2} \right] \times \frac{1}{(1 + g_m R_s)}$$

$$V_0 = -g_m V_{gs} R_D$$

$$\frac{V_0}{V_S} = A_V = \frac{-g_m R_D (R_1 \parallel R_2)}{(R_{si} + (R_1 \parallel R_2))(1 + g_m R_S)}$$

COMMON DRAIN OR SOURCE FOLLOWER AMPLIFIERS

- The next configuration is common drain or source follower amplifier. It is similar to emitter follower or common collector amplifier in BJT.

The circuit is shown below.

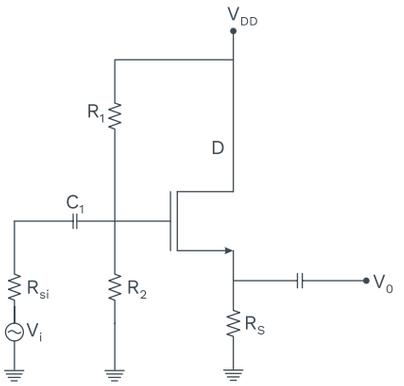


Fig. 5.39 CD Amplifier

DC analysis

- Replace capacitor by open circuit.
- Replace ac source by short circuit.

$$V_G = \frac{V_{DD}R_2}{R_1 + R_2}$$

$$V_S = I_D R_S$$

$$V_{GS} = V_G - V_S$$

$$I_D = \mu_n C_{ox} \frac{W}{2L}(V_{GS} - V_T)^2$$

Solve quadratic equation to find the drain current I_D.

$$g_m = 2\sqrt{KI_D}$$

AC analysis

- Replace capacitor by short circuit.
- Replace dc source by short circuit.

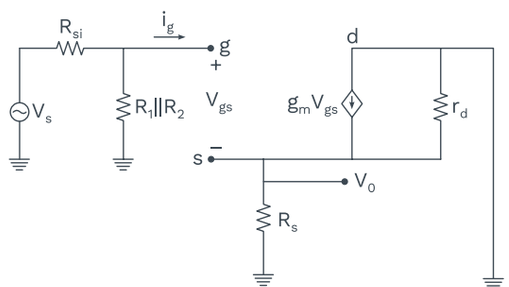


Fig. 5.40 Small-Signal Model of Common Drain Amplifier

- Input resistance
  $R_{in} = R_1 \| R_2$
- Voltage gain
  r_d and R_s are actually connected in parallel; therefore, the output voltage is
  $V_0 = g_m V_{gs} (r_d \| R_s)$

$$V_{gs} = V_S \cdot \left[\frac{R_1 \| R_2}{R_{si} + (R_1 \| R_2)}\right] = \frac{1}{1 + g_m (r_d \| R_s)}$$

Substitute V_{gs} in V_0, we get

$$A_v = \frac{V_0}{V_s} = \left[\frac{R_1 \| R_2}{R_{si} + (R_1 \| R_2)}\right] = \frac{g_m (r_d \| R_S)}{(1 + g_m (r_d \| R_s))}$$

- Output resistance
  Short circuit V_s

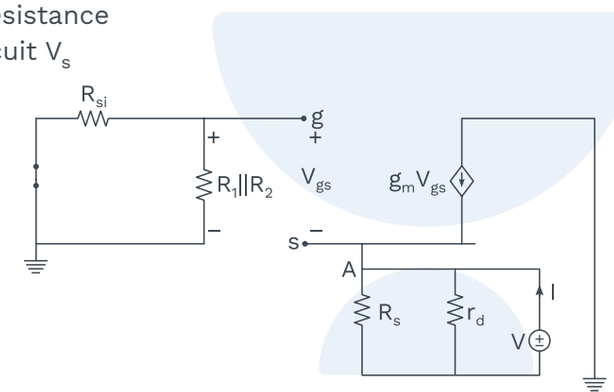


Fig. 5.41 Output Resistance of CD Amplifer

$V_{in} = 0 = (V_{gs} + V)$
$V_{gs} = -V$
Applying KCL at node A, we get

$$I + g_m V_{gs} = \frac{V}{(R_s \| r_d)}$$

$$I - g_m V = \frac{V}{(r_d \| R_s)}$$

$$I = V\left[g_m + \frac{1}{r_d \| R_s}\right] = V\left[g_m + \frac{1}{r_d} + \frac{1}{R_s}\right]$$

$$\frac{V}{I} = \frac{1}{g_m + \frac{1}{r_d} + \frac{1}{R_s}}$$

$$R_{out} = \frac{V}{I} = \left(\frac{1}{g_m} \| R_s \| r_d\right)$$

COMMON GATE AMPLIFIER

- The last configuration to be studied is common gate amplifier.
- It is similar to common base amplifier of BJT.
- The circuit for common gate amplifier is shown below.

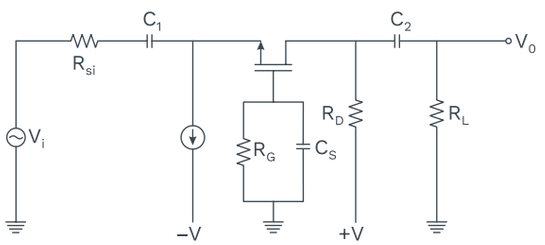


Fig. 5.42 Common Gate Amplifier Circuit

- The small-signal model for the above circuit is shown below.

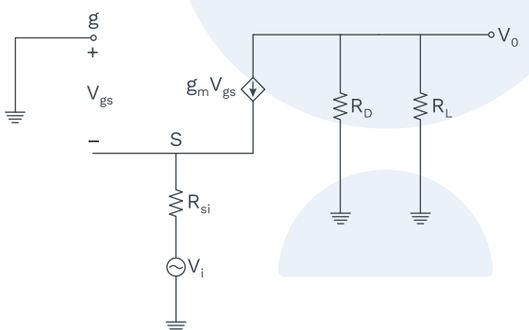


Fig. 5.43 CG Small-Signal Model

Voltage gain

Output voltage $V_0 = -g_m V_{gs}(R_D \| R_L)$

$V_S = (V_i + g_m V_{gs} R_{si})$

$V_g = 0$

$V_{gs} = V_g - V_s = -V_i - g_m V_{gs} R_{si}$

$$V_{gs} = -\frac{V_i}{1 + g_m R_{si}}$$

Substitute V_{gs} in V_0, we get

$$A_v = \frac{V_0}{V_i} = \frac{g_m(R_d \| R_L)}{1 + g_m R_{si}}$$

Input resistance

$$R_i = \frac{V_i}{I_i} \qquad \text{(ignore } R_{si}\text{)}$$

$$= \frac{-V_{gs}(1 + g_m R_{si})}{-g_m V_{gs}}$$

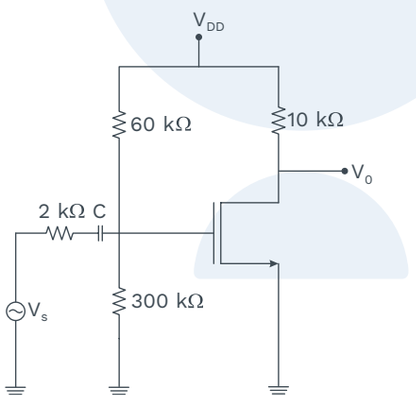
$$R_i = \frac{1}{g_m} + R_{si}$$

Output resistance

$R_{out} = R_D$

Example 13:

In the circuit shown below, the parameters are g_m = 1 mA/V, r_0 = 50 kΩ. What is the value of gain?



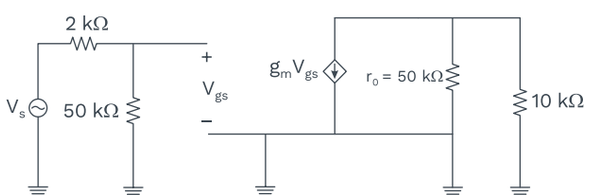
Sol:

The small-signal model for the above circuit is shown below:

$R_G = 60||300$

$$R_G = \frac{60 \times 300}{60 + 300}$$

R_G = 50 kΩ



$$V_{gs} = V_s \times \left(\frac{50}{50+2}\right)$$

$$V_0 = -g_m V_{gs}(r_0 || 10)$$

$$\frac{V_0}{V_s} = -1\text{mA/V} \times \frac{50}{6}\text{k} \times \frac{50}{52}$$

$$\frac{V_0}{V_s} = \frac{-2500}{312} = -8.012$$

Chapter Summary



- JFET is a three terminal device.
- FET is voltage controlled, whereas BJT is a current controlled device.
- FET has various advantages over BJT such as high input impedance, low noise, good thermal stability etc.
- FET is a unipolar device, whereas BJT is a bipolar device.
- The operation of FET depends on majority or minority charge carriers but not on both.
- The JFET is nothing but a voltage controlled resistor.
- The JFET is classified as ann-channel and p-channel.
- The maximum current of JFET is I_{DSS}, and it occurs when V_{GS} = 0 V.
- The pinch-off voltage is the voltage, at which the channel is pinch-off and the current is zero.
- In FET devices, the gate to source voltage V_{GS} actually controls the drain current I_D.
- The relationship between drain current (I_D) and gate to source voltage (V_{GS}) is non-linear and is given by Shockley's equation.
- The 3 types of biasing used in JFET are
  - **i)** Fixed bias circuit
  - **ii)** Self bias circuit
  - **iii)** Voltage divider bias circuit.
- Transconductance is the variation of drain current (I_D) with gate to source voltage)V_{GS}). It is denoted by g_m.
- MOSFET stands for Metal Oxide Ssemiconductor Field Effect Transistor.
- The process of variation of channel width by variation of V_{DS} is called channel width modulation.
- MOSFETs has input impedance greater than JFET.
- MOSFETs has various advantages over JFET.
- MOSFETs are faster than JFETs.
- MOSFETs is nothing but a capacitor, in which we can control voltage.
- MOSFETs is available in two types depletion and enhancement.

Chapter Summary



- The depletion and enhancement type: MOSFETs are further classified into n-channel and p-channel.
- The characteristics of depletion type MOSFET is the same as that of JFET.
- The direction of arrow in the symbol of n-channel JFETs or MOSFETs will always point to the center of the symbol, whereas for the p-channel device, it will point out of the center of the symbol.
- MOSFET can be used as an amplifier in the saturation region.
- The three regions of operation of MOSFET are triode, saturation and cut-off.
- The analysis of p-channel FETs is the same as that of n-channel FETs except for the fact that all the voltages will have opposite polarity, and the current direction will be opposite.
- Channel conductance is the inverse of channel resistance and is denoted by g_m.
- The three main configurations, in which MOSFET amplifier work is:
  - **i)** Common source amplifier
  - **ii)** Common gate amplifier
  - **iii)** Common drain amplifier

PRACTICE QUESTIONS

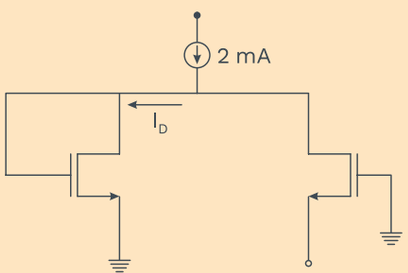
Q1 **In the figure shown below, both the MOSFETs are identical, and these current-voltage characteristics are given by:**

$V_{DS} < V_{GS} - V_T$

$$I_D = \left[\left(V_{GS} - 1\right)V_{DS} - \frac{V_{DS}^2}{2}\right] \text{mA}$$

$V_{DS} \geq V_{GS} - V_T$

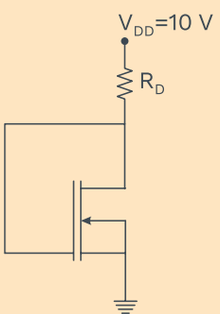
$I_D = (V_{GS} - 1)^2$ mA



The value of current I_D is:

a) 1 mA **b) 2 mA** **c) 3 mA** **d) 4 mA**

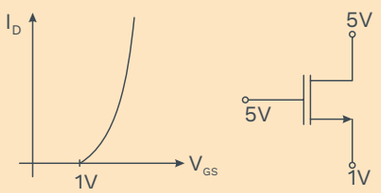
Q2 **Given an n-MOS circuit as shown in the figure, the specification of the circuit includes** $V_{DD} = 10\text{ V}, k = \mu_n C_{ox}\frac{W}{L} = 10^{-4}\text{ A/V}^2, V_T = 1\text{ V}$ **and** $I_{DS} = 0.5$ **mA. The value of** V_{DS} **and** R_D **are respectively:**



a) 8.32 V, 11.68 kΩ **b) 4.16 V, 5.84 kΩ**

c) 4.16 V, 11.68 kΩ **d) 8.32 V, 5.84 kΩ**

Q3 **For an n-channel MOSFET, the transfer characteristics and its connection are shown in the figure:**

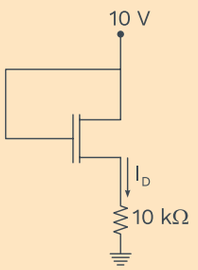


Find the threshold voltage and region of operation, respectively.

a) 1 V, Active **b) 1 V, Satruation**

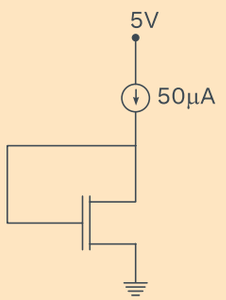
c) 1 V, Triode **d) 1 V, Cut-off Region**

Q4 **The parameters for a given transistor are V_{TN} = 2 V and K_n = 0.2 $\mu A/V^2$**
The power dissipated in (mW) is _____ .



a) 0.61 **b) 0.488** **c) 0.244** **d) 0.122**

Q5 **The parameters of transistor shown below are V_T = 1.2 V and K_n = 0.5 mA/V^2 and λ = 0. The voltage V_{DS} is _____.**

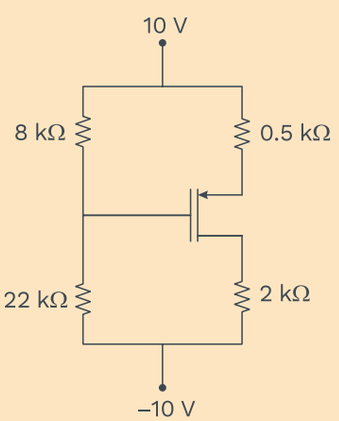


a) 2.4 V **b) 0.5 V** **c) 1.516 V** **d) 0.78 V**

Q6 **In the circuit shown below-**

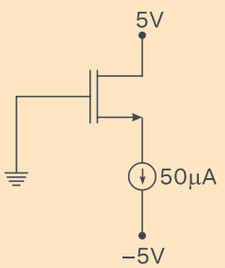
V_{TP} = –2 V, K_p = 1 mA/V^2

The value of drain current I_D is _____ mA.



a) 3.12 **b) 6.24** **c) 1.56** **d) 5.76**

Q7 **The parameters of transistor shown below are V_{TN} = 1.2 V, K_n = 0.5 mA/V^2 and λ = 0. Find V_{DS}?**

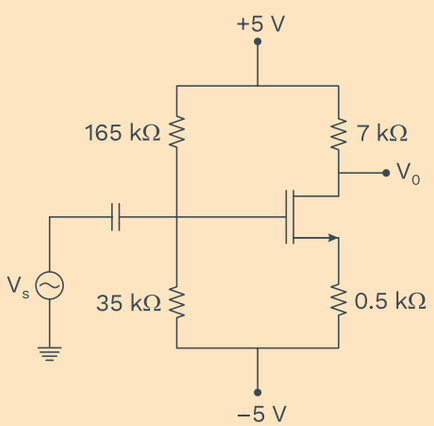


a) 4.516 V **b) 5.516 V** **c) 6.516 V** **d) 7.516 V**

Q8 **To operate MOSFET as an amplifier circuit, which of the following region is useful?**

a) Linear **b) Triode** **c) Cut-off** **d) Saturation**

Q9 **Consider the CS circuit shown below; the transistor parameters are V_{TN} = 0.8 V, K_n = 1 mA/V^2 and λ = 0. The small-signal voltage gain is:**



a) −10.76 **b) −8.76** **c) −7.76** **d) −5.76**

Q10 **If the pinch-off voltage of JFET is 6V, then the cut-off voltage is ____.**
a) 3 V **b) 4 V** **c) 5 V** **d) 6 V**

Answer key:

1) a)	**2) c)**	**3) b)**	**4) d)**
5) c)	**6) a)**	**7) c)**	**8) d)**
9) d)	**10) d)**		